PUBLICATION PROPOSAL · DISCUSSION DRAFT · 29 OCTOBER 2026

The battery as seen from flight: LiPo versus Li-ion on a real quadrotor, with end-of-discharge ground truth

Battery model identification from flight logs and remaining-flight-time error, measured against real discharges to cutoff, comparing PX4's stock estimator with an equivalent-circuit model.

Kevin Medrano (TIDOP, University of Salamanca) · collaboration with the energy group · platform RJX F450 / PX4 1.17 / RTK

One-sentence idea. Consumption models and "energy-aware" control are saturated in simulation. What nobody publishes is the field measurement with ground truth: how wrong the remaining flight time shown to the pilot is, per battery chemistry, when the pack is actually discharged to cutoff. This drone has already lived that failure (0 % reported at a real 24.1 V) and has everything needed to measure it.



RJX F450 carbon frame, 450 mm



4 × T-Motor F90 1300 KV, 45 A peak each



LiPo 6S 1300 mAh 100C (current, 4.8 min of flight)



NxtPX4 v2 FC, custom PX4 1.17 build, bidirectional DShot

4.8 min
real flight on 1300 mAh down to 3.60 V/cell (log 18)

2 mm
static RTK-fixed scatter (91/91 samples)

0 %
battery level PX4 reported at a real 24.1 V (27 Oct)

~3 C
mean drone current on a 4.5 Ah pack

What we ask of the energy group

Battery bench

OCV-SoC curve, HPPC pulses at 2-3 temperatures and a reference capacity test of each pack every ~10 flights.

Model and estimator

Equivalent circuit (1-2 RC + R0(SoC,T)) identified from flight and from the bench; SoC estimator with a Kalman filter (UKF).

Interpretation

Why each chemistry behaves as it does at 3-15 C, what the voltage cutoff leaves unused, and what to recommend to the autopilot.

1 · Why: the gap in the literature and the real problem

⚠ Gap 1 · control

The savings claimed for energy-aware control (NMPC 36 %, PID tuning 70 %, battery-aware planning 7.5 %) are **all simulated**. There is no real-flight measurement of the energy cost of the controller.

⚠ Gap 2 · battery chosen

Nobody identifies the battery model **from flight logs** at 3-15 C, nor validates **remaining flight time against real discharges to cutoff**. Cell datasets cover 0.1 to 5 C. A thesis on 162 public PX4 logs admits it: "no ground truth for remaining flight time exists".

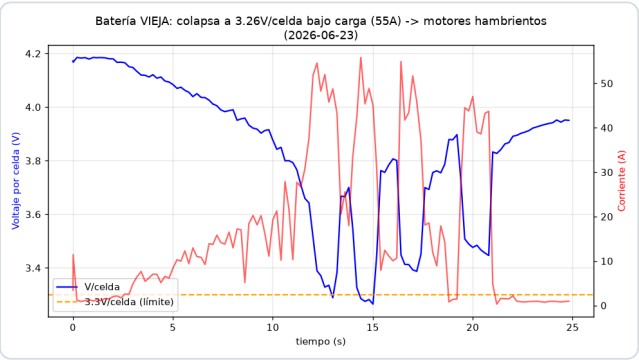
⚠ Gap 3 · dataset

The only open multirotor dataset with current is the 2021 DJI M100 one (closed autopilot). None exists with an open autopilot, calibrated current, RTK, per-motor RPM and pack temperature.

Closest work (and what it leaves open)

Work	What it does	What it does not do
Motion-Specific Battery Health Assessment for Quadrotors (ICRA 2026)	Measures current in flight (S500 + Pixhawk 6C, 4S 2.2 Ah) and replays it in an electrochemical model.	Degradation is simulated; no validation on the pack; number of flights not stated; no temperature, one chemistry.
Dibbern et al., JATE 2023	ECM + EKF/UKF/coulomb counting with mission profiles. Finds that end-of-discharge error does not track SoC error.	Bench data, not flight.
ML-based SoH for UAVs (arXiv 2607.06791, 2026)	342 + 289 flights, capacity measured every 10 flights.	No temperature, no physical model, one chemistry.
Battery-aware predictive planning (arXiv 2609.12188, 2026)	Reduced-order battery model inside the planner.	Simulation only; acknowledges that the controller changes the energy-accuracy trade-off but does not measure it.
LiPo vs Li-ion guides (manufacturers, 2025-2026)	"30-50 % more endurance with Li-ion".	No controlled test on the same drone and the same mission.

The problem, measured on this drone



Real measurement (23 Jun 2026, old pack): cell voltage drops from 4.2 to 3.26 V under 55 A peaks and recovers to 3.95 V when throttle is released. That drop is internal resistance, and it fools any estimator that looks only at voltage. (Plot labels in Spanish: V/celda = V per cell, Corriente = current, tiempo = time.)

27 Oct 2026, field flight. PX4 reported **0 % at a real 24.1 V** (4.0 V/cell, ~75 %): coulomb counting was integrating a bogus current from the ESC. After switching to voltage-based estimation, the percentage swings with the sag under load.

What we want to answer:

- How wrong is the remaining time (in seconds) 1, 2 and 3 min before cutoff, per estimator and per chemistry?
- Does the internal resistance identified *in flight* match the bench value (HPPC)?
- How much energy does the 3.5 V/cell cutoff under load leave unused in Li-ion versus LiPo?
- Is setting `BAT1_R_INTERNAL` per chemistry in PX4 enough to fix it?

2 · What is tested: two matched chemistries, three estimators

LiPo pouch, high discharge

Pack	**6S1P 4500 mAh 60C** (×2)
Nominal energy	~100 Wh
Mass	~510 g
Continuous discharge	270 A
Internal resistance	low (mΩ per cell, to be measured)
Typical use	FPV, small industrial

Li-ion cylindrical 21700, high discharge

Pack	**6S1P Molicel P50B**, 5000 mAh (×2)
Nominal energy	~108 Wh
Mass	6 × 70 g + assembly ≈ 460 g
Continuous discharge	60 A (45 A if P45B)
Internal resistance	several times the LiPo's (to be measured)
Typical use	long range, industrial endurance

Same voltage, same nominal capacity and same mass (±10 %): the only thing that changes is the chemistry. Two packs of each to separate pack-to-pack variation. Semi-solid state ruled out (1-3 C continuous, a 4.5 Ah pack cannot sustain this drone); LiHV is not a new chemistry; LiFePO4 is not used in multirotors.

Hypotheses (stated before measuring)

H1. Li-ion delivers more Wh per gram, but its higher internal resistance makes the 3.5 V/cell cutoff *under load* leave more energy unused than in LiPo.

H2. PX4's voltage-based estimator (the stock one) is more wrong on Li-ion than on LiPo; coulomb counting with calibrated current is equally wrong on both; ECM+UKF is the best on both.

H3. R0 identified from flight logs matches the bench (HPPC) within 15 %, and setting it in `BAT1_R_INTERNAL` measurably reduces PX4's remaining-time error.

The three estimators compared (all on the same flights, by offline replay)

Estimator	Input	What happens to it in flight	Who uses it
PX4 voltage-based (stock)	Pack voltage + optional IR compensation (`BAT1_R_INTERNAL`)	Swings with the sag; with high internal resistance it badly underestimates under load.	Thousands of PX4 drones
Coulomb counting	Calibrated current integrated over capacity	Good if the real capacity is up to date; blind to temperature and to energy not extractable at high current.	PX4 optional, DJI, BMS
ECM + UKF (proposed)	Voltage, current, temperature, equivalent-circuit parameters	Separates OCV from the IR drop; predicts end of discharge under the real load.	Battery literature; here, in flight

3 · How it is measured: instrumentation on the drone

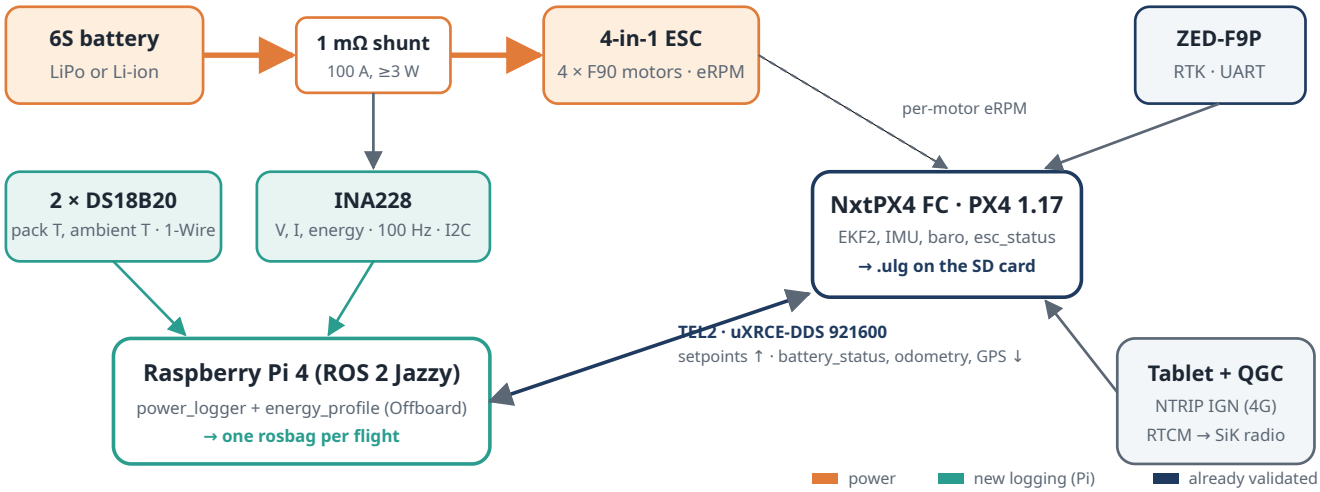
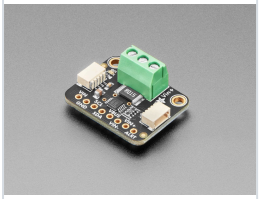


Figure 1. Two independent records of the same flight: the autopilot writes its log to the SD card and the Pi records current, voltage and temperature through its own path. They are aligned afterwards using the voltage, which both sensors see. The ESC current is not used (it is invalid: it reads 13.6 A with the drone sitting still).

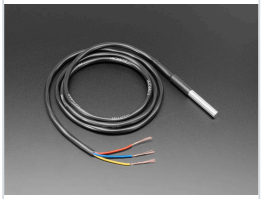
What gets installed (product photos)



INA228 I2C breakout, 20-bit. Used with an external shunt (the on-board one is 15 mΩ, ≤10 A).



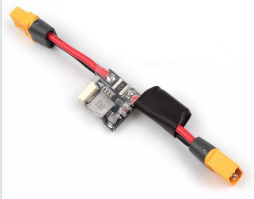
External shunt on the XT60 positive lead. 100 A → 100 mV; dissipates 0.4 W at 20 A.



DS18B20 encapsulated, 1-Wire. One taped to the pack, one for ambient.



Molicel INR21700-P50B: 5 Ah, 60 A, ~70 g. Six in series = the Li-ion pack.



Phase 2: analog power module (Holybro PM02 or Mauch Hall) into the FC's current pad.

Product images: Adafruit (INA228, DS18B20), 18650batterystore (P50B), PX4 documentation (PM02). Illustrative only; the exact part is chosen at purchase.

What the Pi measures (new)

- INA228 + external shunt: V, I at 100 Hz and energy integrated on-chip.
- Pack and ambient temperature at 1 Hz.
- Profile phase (takeoff, hold, up, down, out, back, land).
- Optional: per-cell voltage with an ADS1115 on the balance lead.

What the FC measures (already)

- Pack voltage (calibrated to <0.5 %).
- eRPM of the 4 motors via bidirectional DShot.
- IMU, barometer, attitude, flight mode.
- RTK position and velocity (cm) and fix type.
- Battery log rate raised from 5 to 100 Hz by a file on the SD card.

What the bench measures (energy)

- OCV versus SoC per chemistry.
- HPPC: R0 and RC constants at 2-3 temperatures.
- Reference capacity every ~10 flights (fade).
- Per-cell resistance with an ESR meter.

4 · How it is flown: three identical profiles, Offboard with RTK

The drone flies autonomously, commanded from the Pi, with the same setpoints on every flight. RTK (2 mm static, cm in flight) makes "the same setpoint" the same physical place. Each flight ends **on battery, not on time**: when the filtered cell voltage drops below 3.5 V under load, the drone returns to the mark and lands. That is the ground truth for remaining time.

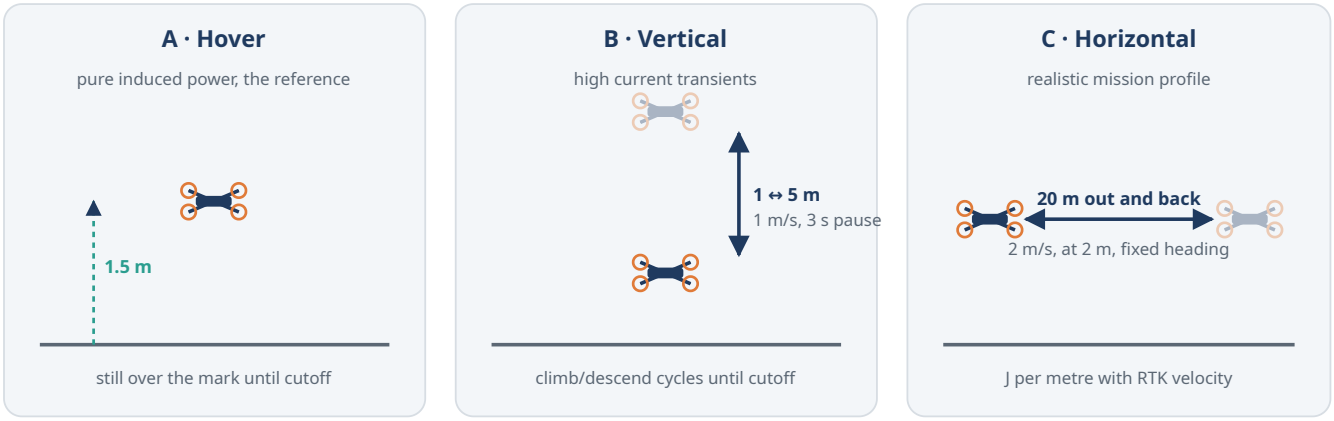


Figure 2. Profiles relative to the takeoff mark, with a fixed compass heading. Every flight: charge to 4.20 V/cell, 30 min rest, weigh the drone, log wind and temperature, fly to cutoff.

Sequence of one flight

1. Pack at ambient temperature, charged to 4.20 V/cell and rested 30 min.
2. Drone on the mark; RTK fixed required to take off (node gate).
3. Pilot sets the switch to Offboard and arms; the node takes off and runs the profile.
4. If the fix drops to float in flight, the flight continues and the segment is tagged; no abort.
5. Filtered cell voltage (2 s) < 3.5 V under load → return to the mark and land.
6. 30 min rest and final OCV reading; record in the session manifest.

Experimental matrix

Factor	Levels
Chemistry	LiPo 4500 60C · Li-ion P50B (2 packs each)
Profile	Hover · Vertical · Horizontal
Repetitions	≥ 5 per cell, ABBA order within a session
Total	**30-40 flights to cutoff**, 4-6 field sessions
Duration per flight	~10-12 min (log 18 scaled to the 4.5 Ah pack)

Safety with 1P Li-ion: cap PX4's maximum thrust at ~0.6 (plenty: 9.4 kg of thrust for ~1.5 kg) and gentle profiles. Pilot on RC as backup at all times.

5 · What comes out of the data: analysis and planned figures



All offline and reproducible: the same flights feed the three estimators and allow sweeping PX4 parameters without flying again.

Paper figures (what we want each one to say)

#	Figure	Message
F1	Platform and instrumentation	What is measured, with what, and the two logging paths.
F2	RTK trajectories of the three profiles, all flights overlaid	The profiles are repeatable at cm level: the variable is the battery, not the flight.
F3	V, I and T of a full discharge, LiPo and Li-ion side by side	The same mission produces different curves: sag, heating and cutoff point.
F4	R0 identified in flight versus bench HPPC, by SoC and temperature	Can the battery be characterised from logs without a lab? (H3)
F5	Remaining-time error versus horizon, by estimator and chemistry	The central figure: how wrong the pilot is and with which estimator they stop being wrong. (H2)
F6	Usable Wh in flight versus bench Wh, by chemistry and profile	What the voltage cutoff leaves unused. (H1)
F7	Endurance in minutes and in minutes per gram	The number everybody quotes without measuring, measured.
F8	Reference capacity over the campaign	Fade over 30-40 flights; reported, not promised.

Metrics

Battery

- SoC error (RMS, max)
- Remaining-time error in s at 1, 2 and 3 min before cutoff
- R0, τ1, τ2 flight vs bench
- Usable Wh / bench Wh

Flight

- Mean power per phase
- J per metre (profile C) with RTK velocity
- Hover power versus mass, against momentum theory
- Pack temperature rise

Quality

- RTK scatter per profile
- Wind per flight (anemometer)
- % of RTK-fixed samples
- Attitude tracking error

6 · Plan, roles and where to publish

Drone side (TIDOP)

- Instrumentation: INA228 + shunt, DS18B20, clamp-meter calibration.
- Pi nodes: power logging and Offboard profiles with battery-based termination.
- Field campaign with pilot and RTK; logs and rosbags.
- Synchronisation, phase segmentation, replay of PX4's estimator.
- Open dataset (Zenodo) and flight figures.

Energy side

- Bench protocol: OCV-SoC, HPPC at 2-3 temperatures, reference capacity, per-cell ESR.
- Equivalent-circuit model and identification (bench and flight).
- ECM + UKF estimator and end-of-discharge error analysis.
- Electrochemical interpretation of the LiPo / Li-ion differences.
- Assembly of the 6S1P Li-ion pack if not bought ready-made.

Timeline (from material arrival)

Weeks	Milestone	Status
0	Purchase: 2 LiPo 4500 60C, 2 Li-ion P50B with balance lead, INA228, 1 mΩ 100 A shunt, 2 DS18B20, DC clamp meter, anemometer.	pending
1-2	Instrumentation on the drone, Pi nodes, bench calibration without propellers, test 0: outdoor Offboard hover with RTK.	pending
2-3	Bench tests of the four packs (OCV, HPPC, capacity).	energy
3-6	Campaign: 4-6 sessions, 30-40 flights to cutoff.	field
6-9	Identification, estimators, figures.	both
9-12	Writing and submission.	both

Already validated on the drone (nothing to invent)

- ok Offboard link over DDS with PX4 1.17.
- ok RTK fixed with IGN NTRIP from the tablet (flights 17/18: 92-97 % fixed).
- ok Compass calibrated, outdoor profile, failsafes and RC override.
- ok FC voltage calibrated; per-motor eRPM in the log.
- ok Outdoor takeoff/hold/land node (still to be flown).
- no Current: the ESC channel is invalid. Hence the INA228.

Venues and risks

Venues: Journal of Energy Storage or Energies (battery angle); Drones or Aerospace Science and Technology (UAV angle); IEEE Trans. Transportation Electrification. The dataset, separately, in Scientific Data or Data in Brief.

Risks: (1) "LiPo ECMs are already studied": the defence is the 3-15 C regime, identification from flight, the ground truth and two chemistries on the same drone. (2) Small fade over 30-40 flights: reported, SoH not promised. (3) Wind: anemometer per flight, discard gusts, ABBA order.

Second paper with the same instrumentation: energy per mission of the controller (PID, jerk-limited and MPC, already implemented on the Pi) with the accuracy-energy Pareto front measured in the field; the published savings (7-70 %) are all simulated.

Key references

Motion-Specific Battery Health Assessment for Quadrotors, ICRA 2026 · arxiv.org/abs/2603.12791

Battery-Aware Predictive Trajectory Planning for Multirotors, 2026 · arxiv.org/abs/2609.12188

Rodrigues et al., DJI M100 energy dataset, Sci. Data 2021 · Bills et al., eVTOL battery dataset, Sci. Data 2023

Dibbern et al., load-dependent SoC and end of discharge of a multirotor battery, JATE 12(2), 2023

ML-based battery SoH prediction for UAV maintenance, 2026 · arxiv.org/abs/2607.06791

Full review and decisions (Spanish): docs/publicaciones/2026-10-29_paper_energia_bateria_campo.md (px4_drone_dev repo)